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ISP Training Program – VSS on Jetson Thor

Chun-Yu Chen,

Solution Application Engineer, Advantech

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Agenda

- Brief Introduction to VSS on Jetson Thor
- Tips and Pitfalls for Using VSS
- Examples
  - Video File as Source: Warehouse Example
  - Camera Stream as Source: Clearance Space Detection
- Exercises

Course Material: https://github.com/chunyu1021/2026-ISP-Global-Training-Program

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ISP Training Program

Training Date	Projects	Hardware	NVIDIA Base	GitHub
2026.04.15 (en) 2026.04.22 (zh)	Edge Agent	MIC-733	Agent Studio	LINK
2026.04.29 (en) 2026.05.06 (zh)	VSS on Jetson Thor	MIC-742	VSS 2.4.1	LINK
TBD	Perceptor/ReMEmbR	MIC-732	Issac Perceptor/ReMEmbR	Perceptor/ReMEmbR
TBD	AMR SLAM Navigator	MIC-732	-	LINK
TBD	Omniverse/Issac Sim	-	-	-

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Brief Introduction to VSS on Jetson Thor

VSS on Jetson Thor

- VSS = Video Search and Summarization
  - Supports video/image file upload and live camera stream as inputs
  - Summarization, Q&A, and alerts on files and live streams
- Based on NVIDIA's VSS 2.4.1
  - Originally, only alert function can fully run on Jetson Thor
  - Hybrid deployment
    - VLM and CV pipeline are executed locally on Jetson Thor
    - LLM, Embedding Model, Reranker Model, and Audio processing are hosted remotely on server
- ***Advantech unlocks full functions on MIC-742***
  - All components (models, database, backend) are deployed on MIC-742

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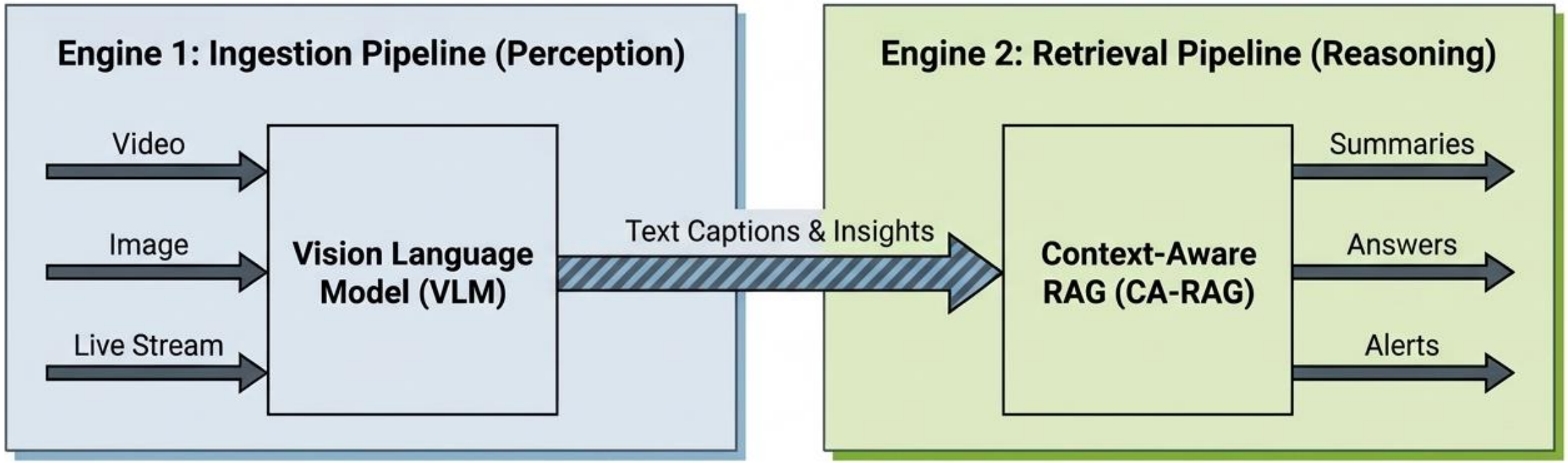
Architecture: Ingestion Pipeline & Retrieval Pipeline



- ***Ingestion Pipeline***: Extract visual insights from the input video in the form of captions and scene descriptions.
- ***Retrieval Pipeline***: Visual insights from the ingestion pipeline are further processed, indexed, and used in retrieval tasks like summarization and Q&A.

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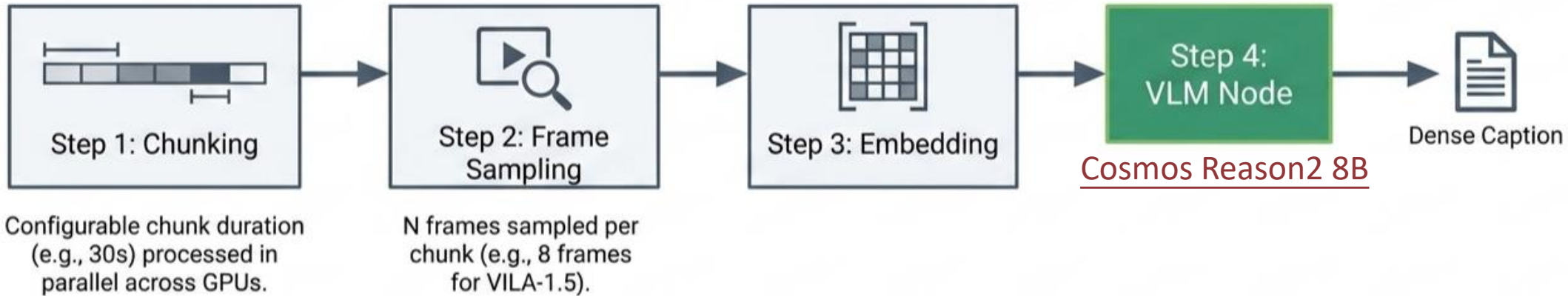
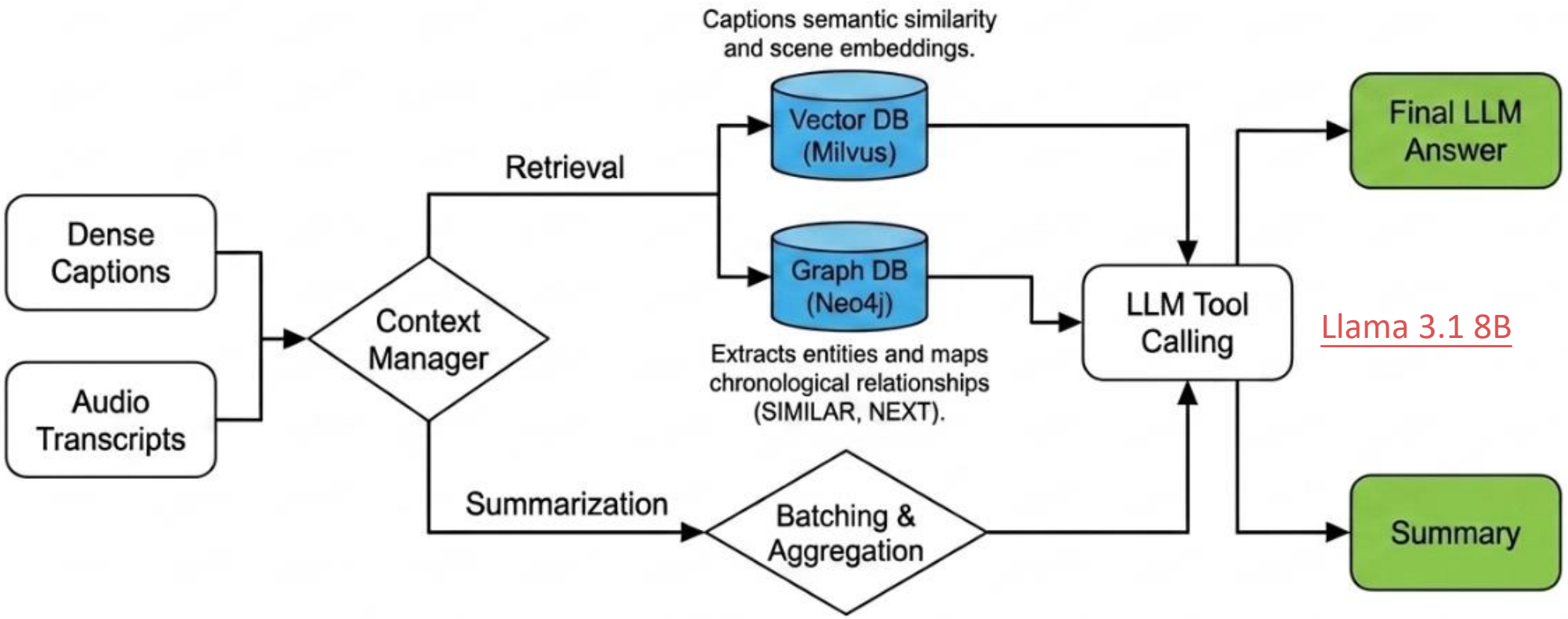
Ingestion Pipeline



- Video files are processed as small chunks (typically a few seconds or a few minutes, based on model and use case).
- The frames from each chunk are sampled to select a limited number of frames from the entire chunk.
- VLM captions are generated using the embeddings and user prompt for each chunk.

Retrieval Pipeline



- Retrieval pipeline, where CA-RAG plays a major role, is responsible for processing the output of the ingestion pipeline.
- ***Summarization*** and ***Data Retrieval (Q&A)*** used different workflows.

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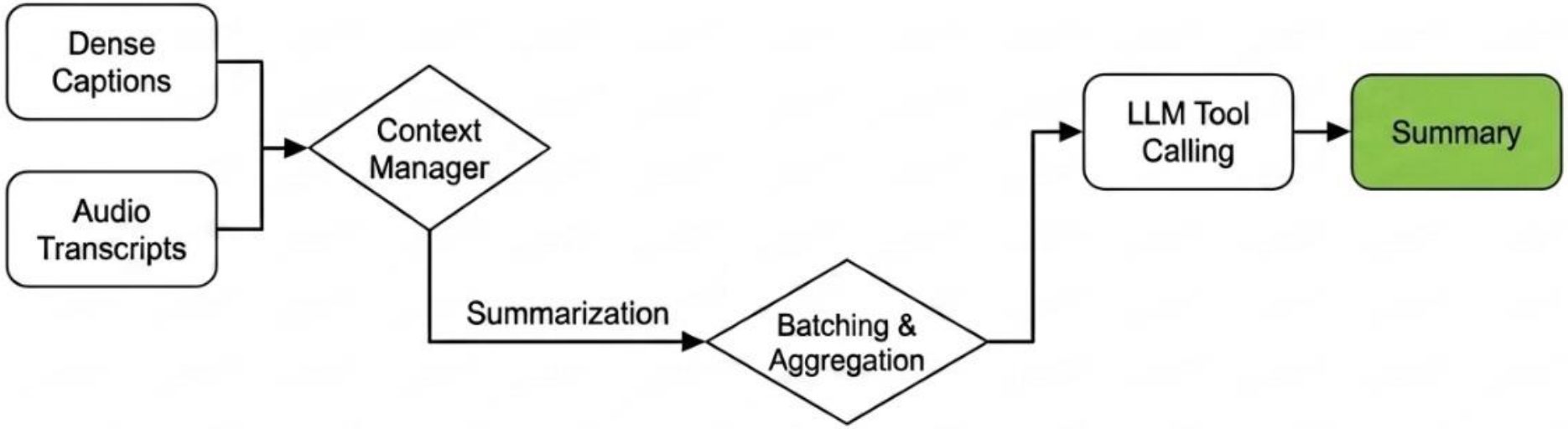
Summarization



- The ***caption prompt*** is used in Ingestion Pipeline for the VLM to focus on keywords or events in the chunks.
- Two-stage processing to generate summary
  - Batching: Groups together captions into batches and generates summaries for each batch. Summary is generated by using a ***caption summarization prompt***.
  - Aggregation: Combines batch summaries using a ***summary aggregation prompt***.

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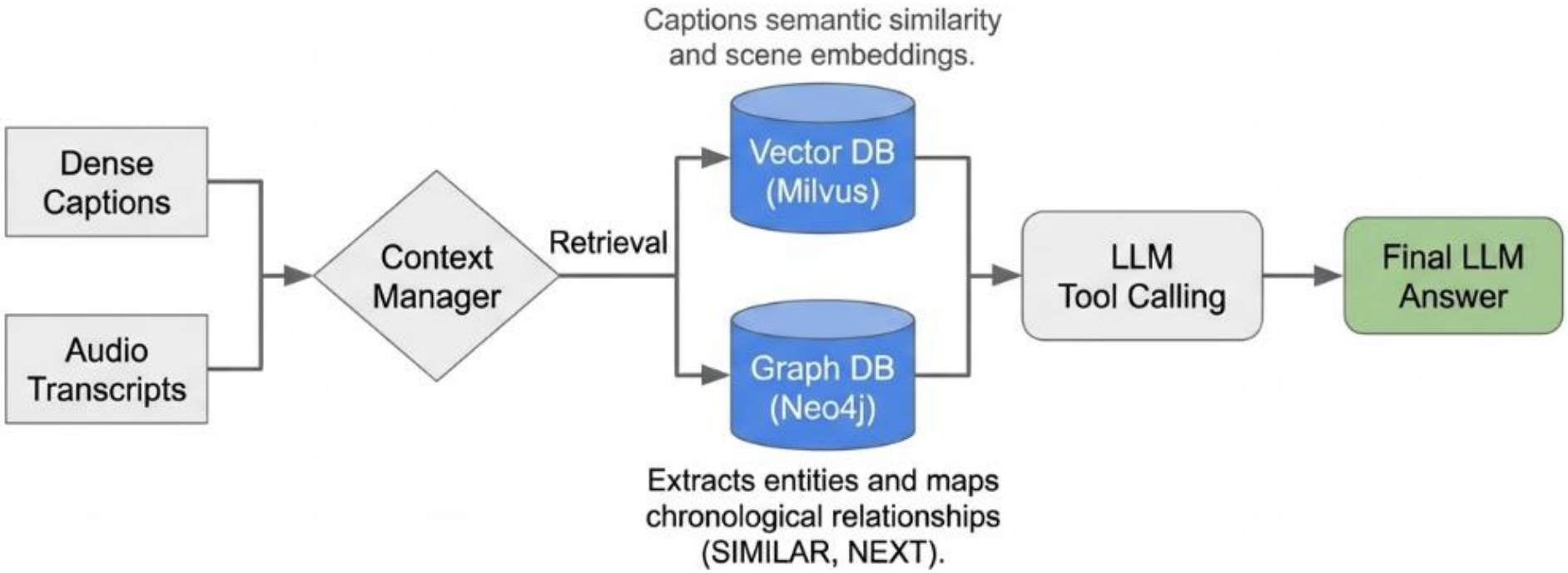
Data Retrieval (Q&A)



- Vector RAG utilizes a vector database to store caption embeddings, retrieving the most ***semantically similar*** video chunks to answer a query.
- Graph RAG utilizes a graph database to store extracted ***entities and their relationships***, navigating this knowledge graph to retrieve complex interactions and provide context-aware answers.

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Tips and Pitfalls for Using VSS

Before Installation…

- Prepare MIC-742 with BSP version 7.1
- Follow installation guide to set up Docker
  - Docker Installation Guide
- Connect your device with internet
- Generate Hugging Face access token (documentation)
  - The token we need here is the one with Read permission
  - No needs to use Fine-grained or Write permission token
- Get access to the model Cosmos-Reason2-8B (documentation)

First-Time VSS Setup

- Disable offline mode (send the commands in terminal)
  - `export HF_HUB_OFFLINE=0`
  - `export TRANSFORMERS_OFFLINE=0`
- Specify your Hugging Face token in .env file
  - `export HF_TOKEN=<your_hugging_face_token>`
- Use `docker compose up` to launch VSS
- It will take about 5 - 6 minutes to start VSS
- The user interface of VSS will be running at http://localhost:9100

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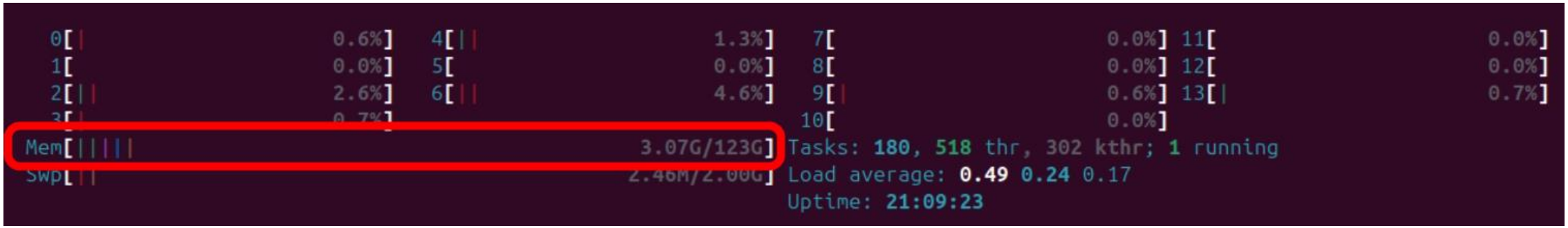
Memory Allocation Issues

- VLLM_GPU_MEMORY_UTILIZATION in .env file preserves memory space for vLLM, which can be adjusted for resolving memory using competition between model and system
- During starting VSS,
  - If you see KV cache memory related errors
    - Increase the value. This gives the model more memory to work with.
  - If you see free memory is less than desired
    - Decrease the value. This leaves more free space for the system.

Shut Down VSS

- Use `docker compose down` to fully terminate all containers
- Run the script: sys_cache_cleaner.sh
  - The script clears out cache memory
  - Use htop for checking memory usage
    - To install htop: `sudo apt install htop`



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Examples

Warehouse Example

- A default example that you can use after you start VSS
- Use a sample ***video file as source***
- Learn from the sample prompts and prompt tunings
  - Caption prompt
  - Caption summarization prompt
  - Summary aggregation prompt
- Increase sampling frames to capture brief incidents
  - Box dropping
  - Someone passes through caution tape
- Demo video

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Clearance Space Detection

- Use Edge Agent video output as VSS ***live stream input*** (RTSP)
- Edge Agent is used as event monitoring and alert system
- VSS provides in-depth analysis and insights of the event
  - Describe how objects arrive in the zone.
  - How long do objects stay in the zone?
  - Check if objects block the hydrant.
- Demo video

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Exercises

Try Out Yourself!

- Try to run summarization and Q&A with sample videos in VSS
- Sample prompts and questions from NVIDIA's official manual
- Build your own use cases and share with us on GitHub Discussions
- Any questions? Go to the project's GitHub Issues

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